EUROPEAN UNION STEM LABORATORY INITIATIVE

SPHERE: MISSION CONTROL

STUDENT USER MANUAL

DOCUMENT REF // EU_SPHERE_STUDENT_V2

> **Inquiry-Based Science Education (IBSE) Focus:** This manual provides a complete hardware guide and digital workflow to construct a physical thermal envelope (an astronaut spacesuit capsule) and map its real cooling dynamics inside a vacuum analog to a live Newtonian mathematical prediction engine.

1. INTRODUCTION & EXPERIMENT OVERVIEW

During deep-space voyages, spacecraft and astronauts are subjected to thermal extremes. In low Earth orbit (such as on the International Space Station), direct solar irradiation can heat surfaces to **121°C**, while objects in orbital shadows radiate energy away, plunging to **-157°C**. Creating reliable, multi-layer thermal envelopes is critical to preserving astronaut safety.

In this laboratory experiment, you will build a protective "spacesuit" for a simulated astronaut (a 100ml PET container filled with hot water at 80°C). You will drop this unit into an ice-bath chamber (0°C) mimicking the thermal void of space, and use the **SPHERE: Mission Control** dashboard as your live telemetry terminal.

In this experiment, you will compare real temperature measurements with a computer model. By comparing both results, you can investigate how different insulation materials affect heat transfer and discuss why real measurements may differ from predictions.

2. HARDWARE ASSEMBLY GUIDE

To ensure the experiment is economically scalable across diverse classrooms, the hardware relies exclusively on affordable, off-the-shelf industrial and household components:

- **Capsule Body:** 1x 100ml wide-mouth PET food jar.
- **Waterproof Cable Connector:** 1x Industrial PG7 Nylon Cable Gland (mounted securely through a 12.5mm hole drilled in the PET jar cap).
- **Thermal Sensor:** 1x DS18B20 digital waterproof temperature sensor probe.
- **Environmental Chamber:** A large tub or bucket filled with crushed ice and water (maintained at 0°C).

ASSEMBLY STEPS:

1. Thread the temperature sensor wire through the waterproof connector.
2. Screw the PG7 gland locknut onto the inside of the jar cap until firmly sealed.
3. Tighten the external dome nut of the PG7 gland. This compresses the internal neoprene rubber seal around the sensor wire, creating a 100% waterproof seal capable of sustaining submersion in ice water.
4. Verify that the sensor probe is suspended freely in the middle of the jar and does not directly touch the bottom or sides of the PET capsule.

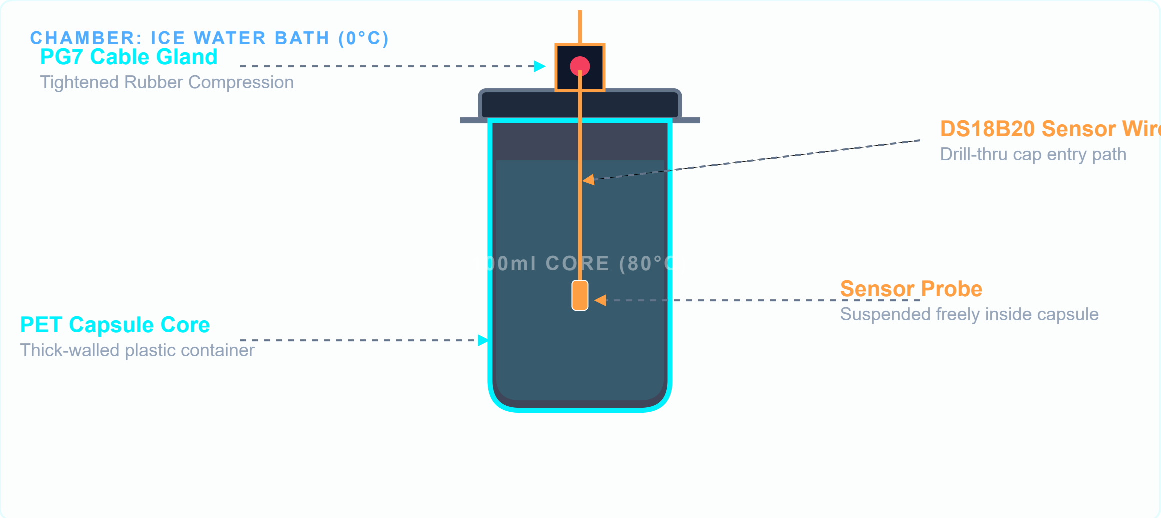

FIGURE 1: SPHERE CAPSULE MECHANICAL ASSEMBLY SCHEMATIC

3. INSULATION CONFIGURATIONS

Your engineering team must design and test different thermal protection systems for the capsule. Each configuration combines materials in a different way to investigate how insulation affects heat transfer.

- **Configuration A — Control Capsule**
  No insulation. This configuration serves as the scientific baseline.
- **Configuration B — Cotton Suit**
  A single cotton layer wrapped around the capsule.
- **Configuration C — Mylar Shield**
  A reflective Mylar foil layer surrounding the capsule. The shiny surface reflects thermal radiation and reduces heat loss.
- **Configuration D — Multi-Layer Thermal Suit**
  A combined insulation system consisting of cotton, bubble wrap, and an outer Mylar layer.

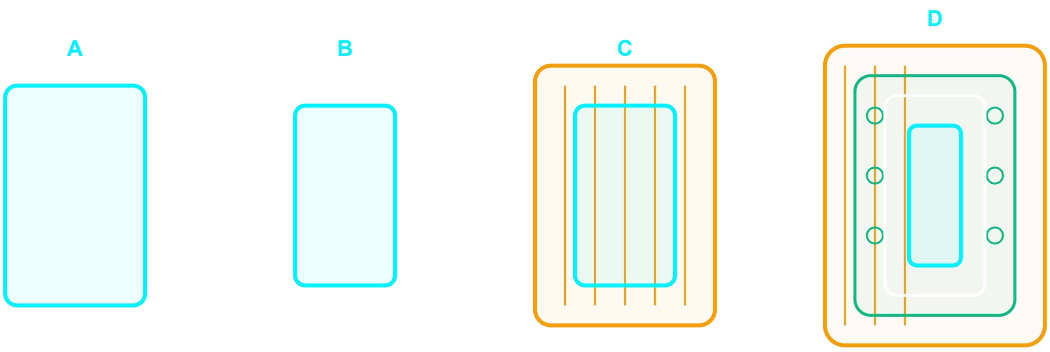

FIGURE 2: INSULATION CONFIGURATIONS TESTED DURING THE EXPERIMENT

4. UNDERSTANDING THE TEMPERATURE GRAPH

As the warm water cools down, heat flows from the warmer capsule to the colder surroundings. By measuring the temperature over time, you can create a graph and compare different insulation materials.

The Mathematics Behind the Model Scientists and Engineers often use a model called Newton's Law of Cooling to describe how objects cool down over time:

$$dT/dt = -k \cdot (T - T_{env})$$
$$T(t) = T_{env} + (T_0 - T_{env}) \cdot e^{-kt}$$

Where:

- **T(t):** The predicted temperature at time `t` (seconds).
- **T_{env} :** The surrounding environmental temperature (typically 0°C for an ice-water bath).
- **T_0:** The initial temperature of the water when the experiment begins (typically 80°C).
- **k:** The cooling decay coefficient (1/seconds) which represents the heat transfer rate. A lower `k` constant indicates a more efficient spacesuit!
- **e:** Euler's mathematical constant (~2.71828).

The simulator uses this model to calculate a prediction curve. You do not need to derive the equation, but you can compare its predictions with your measurements.

Learning Objectives

- Explain how heat flows from warmer objects to colder surroundings.
- Measure and record temperature data at regular intervals.
- Interpret temperature-versus-time graphs.
- Compare the effectiveness of different insulation systems.
- Use a digital model to predict experimental outcomes.
- Compare measured data with simulation results.
- Evaluate why real experiments differ from ideal mathematical models.

5. STEP-BY-STEP MISSION CONTROL WORKFLOW

1. **Unlock Operations:** Read the Mission Briefing on the home page and complete the Pre-Flight Quiz to prove your laboratory readiness.
2. **Run Newtonian Predictions:** Open the **Simulator** tab. Select your insulation type, configure your initial variables ($T_o = 80°C$, $T_{env} = 0°C$), and click **RUN SIMULATION PREDICTION**. Review the projected mathematical decay curve.
3. **Prepare Physical Asset:** Fill the PET capsule with hot water pre-measured at 80°C. Secure the cap and tighten the PG7 gland to prevent convective leakage.
4. **Submerge & Initialize:** Submerge the capsule into the 0°C ice water. Simultaneously click **START LIVE TIMER** in the **Live Telemetry** tab.
5. **Active Sensor Logging:** Every 60 seconds (the alarm terminal will blink green and sound a beep), read the current physical thermometer value and log it into the input box. Click **RECORD DATA POINT** to plot it on the graph overlay.
6. **Analyze Correlation Score:** Once at least 5 points are logged, the telemetry panel automatically calculates your **Thermal Correlation Score**. Discuss with your crew how closely your digital twin simulator aligned with physical reality!

Summer & High-Ambient Temperature Procedures (~30°C):
If you are running this experiment in a hot room (e.g. during summer conditions), note these key steps to ensure high-accuracy and absolute safety:

- **Safety Temp Scaling (Mitigate Scalding Risk):** Instead of using 80°C hot water, you can safely scale the initial temperature down to **55°C–60°C**. Newton's Law of Cooling works identically! Simply enter $T_o = 60°C$ in your simulator prediction, and fill your physical jar with 60°C water.
- **Melting Ice Compensation:** On hot days, your ice bath water may rise slightly above 0°C (e.g. to 4°C). Use your thermometer to measure the actual temperature of your ice bath at the start, and enter that exact value as the environmental temperature T_{env} in your Simulator tab.
- **Ice-Free Tap Water Drop:** If ice is not logistically available, you can drop your capsule into room-temperature tap water (~25°C–30°C). Simply log the tap water's temperature, enter it as T_{env} , and run your predictions!